Preparation memory in the neutral slope

An exact two-step Mehler kernel and its string-time limit

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Abstract

The first neutral coefficient of the actual bilocal retains preparation time even when its charged curvature restriction is stationary. We reduce its selected two-time function exactly to an energy-tilted two-point function plus a positive Gram kernel formed by two q-Mehler transitions. A weighted three-energy estimate gives the compact-time small-deformation expansion with an $O(\varepsilon^2)$ remainder. The additional term is separable at first order and survives the equal mixture that cancels the curvature correction. The complementary odd mixed source produces an order-$\sqrt{\varepsilon}$ energy–slope expectation. This establishes a concrete neutral contribution to the mixed source, not a complete bilocal or bulk particle identification.

1 Objects, actual interface and result

Let h, U be orthogonal unit-variance q-Gaussian fields in the tracial covariant algebra, and put

$$q = e^{-\varepsilon}, \quad \lambda = \varepsilon/4, \quad s = \sqrt{q}, \quad W = \sqrt{\lambda}U, \quad K = h/\sqrt{\varepsilon}, \quad U(u) = e^{iuK}Ue^{-iuK}.$$

The dimensionless string time is $u = 2J_s t$, with $\hbar = 1$. The already established single-selected density [3] is

$$\rho = 1 - s(1+\lambda) + s(h - \sqrt{\lambda}U)^2, \qquad \sigma(Y) = \tau(\rho Y). \tag{1}$$

The normalized actual exact-pinched neutral slope converges to U inside these preparations [4]. Thus the functions below refer to a controlled microscopic coefficient. Take finite-N exact projection, then half-filled fixed-λ double scaling with the original common disorder, then $\varepsilon \downarrow 0$.

Write $g(u) = \operatorname{sech}^2 u$ and $R(u) = \frac{1}{2}g(u)[u\tanh u - 3\tanh^2 u]$.

Theorem 1. *For fixed $S, k < \infty$, uniformly on $|u|, |v| \le S$ including all partial derivatives of total order at most k,*

$$\begin{aligned}\tau(U(u)U(v)) &= g(u-v) - \tfrac{\varepsilon}{2}R(u-v) + O(\varepsilon^2),\\ \sigma(U(u)U(v)) &= g(u-v) + \tfrac{\varepsilon}{2}[R(u-v) + g(u)g(v)] + O(\varepsilon^2).\end{aligned} \tag{2}$$

U has unit *tracial* variance. Its selected variance at the preparation time is $1 + s\lambda(1+q)$, not one. There is no N-uniform rate or growing-time limit in this statement.

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2 Four fields become a classical three-energy chain

Let μ_q be the unit-variance q-Hermite law and P_r the q-Mehler operator,

$$P_r H_n = r^n H_n, \qquad P_r f(a) = \int K_r(a,b) f(b)\, \mu_q(db).$$

Its positivity, trace preservation and semigroup law are standard [1]. Define the probability measure

$$d\Pi_q(a,b,c) = \mu_q(da) K_q(a,b) \mu_q(db) K_q(b,c) \mu_q(dc). \tag{3}$$

The subscript q on K_q denotes its transition parameter, with the same deformation parameter q understood.

Lemma 1. *For bounded continuous f,g,k on the energy spectrum,*

$$\tau[(U^2-1) f(h) U g(h) U k(h)] = (1+q) \int f(a) g(b) k(c)\, d\Pi_q. \tag{4}$$

Proof. It suffices first to use monomials. Label the four U endpoints in cyclic order by $0,1,2,3$, with no h's between 0 and 1. The pairing $(0,1)(2,3)$ is removed by U^2-1. The two remaining pairings are $(0,3)(1,2)$ and $(0,2)(1,3)$, with mutual U-crossing weights 1 and q. In both cases an h-chord joining the f and g blocks crosses one U-chord, one joining g and k crosses one, and one joining f and k crosses two. Within-block weights are unchanged. After summing the h-pairings, both contributions have the q-Gaussian three-block law with covariance

$$\begin{pmatrix} 1 & q & q^2 \\ q & 1 & q \\ q^2 & q & 1 \end{pmatrix}.$$

By the known q-Gaussian Markov property this ordered three-block law is exactly (3). Equivalently, conditional expectation of the third block onto the first two is $P_q k$ of the middle field, leaving $\int (P_q f)(b) g(b) (P_q k)(b)\, \mu_q(db)$. Polynomial approximation extends the identity to the stated class, since h and U are bounded at fixed q. □

This is an application of the established q-Wick rule and q-Ornstein–Uhlenbeck transitions [1, 2]. The two transitions are an auxiliary spectral representation of this contraction; they do not introduce two independent physical disorder samples.

3 The exact selected two-time function

Separate the density as

$$\rho = f_q(h) + s\lambda(U^2 - 1) - s\sqrt{\lambda}\{h, U\}, \qquad f_q(h) = 1 - s + sh^2.$$

The last term is odd under $U \mapsto -U$, whereas $U(u)U(v)$ is even. It therefore vanishes for this observable, but not for arbitrary mixed observables. Let $\widetilde{C}_{1,\varepsilon}$ denote the energy-tilted q-Mehler correlation from [5]. Lemma 1 gives the exact identity

$$S_\varepsilon(u, v) := \sigma(U(u)U(v)) = \widetilde{C}_{1,\varepsilon}(u - v) + s\lambda(1 + q)B_\varepsilon(u, v), \tag{5}$$

$$B_\varepsilon(u, v) = \int e^{i[u(a-b)+v(b-c)]/\sqrt{\varepsilon}}\, d\Pi_q(a, b, c). \tag{6}$$

The trace correlation is $C_{1,\varepsilon}(u - v)$. In the first term of (5), cyclicity and $[h, f_q(h)] = 0$ combine the exterior phases; this reduction would be invalid for the full density.

The extra kernel is positive in its two time indices. Indeed

$$A_u(b) = e^{-iub/\sqrt{\varepsilon}} P_q(e^{iu\,\cdot/\sqrt{\varepsilon}})(b), \qquad B_\varepsilon(u, v) = \int A_u(b)\overline{A_v(b)}\, \mu_q(db).$$

In particular $|A_u| \le 1$ and $|B_\varepsilon| \le 1$. Spectral sign reversal and chain reversal imply reality and symmetry in u,v. The semigroup law gives two useful exact reductions:

$$B_\varepsilon(u, 0) = C_{1,\varepsilon}(u), \qquad B_\varepsilon(u, u) = C_{2,\varepsilon}(u). \tag{7}$$

Consequently

$$S_\varepsilon(u, u) = 1 + s\lambda(1 + q)C_{2,\varepsilon}(u). \tag{8}$$

At fixed $0 < q < 1$, $C_{2,\varepsilon}(u) \to 0$ as $|u| \to \infty$, by the absolutely continuous gap measure and the Riemann–Lebesgue lemma. The selected variance therefore approaches one from its initial value $1 + s\lambda(1 + q)$. This fixed-q statement is separate from the compact-time small-ε expansion.

The one-point function vanishes. Only the anticommutator term could contribute, and its two-energy integral contains $(a + b)e^{iu(b-a)/\sqrt{\varepsilon}} K_q(a, b)\mu_q(da)\mu_q(db)$. Exchange of a,b removes its sine part, and simultaneous sign reversal removes its cosine part. Thus (8) is a variance change, not the square of a nonzero mean.

4 A uniform three-energy estimate

Use angular energies $e_\theta = 2\cos\theta/\sqrt{1-e^{-\varepsilon}}$. For the middle angle and the two successive gaps write

$$\theta_1 = \pi/2 + \sqrt{\varepsilon}x/2, \quad \theta_0 = \theta_1 + \varepsilon z_1, \quad \theta_2 = \theta_1 - \varepsilon z_2.$$

Let $m_\varepsilon(x)$ be the middle marginal density and $p_\varepsilon(y,z) = p_{1,\varepsilon}(y,z)$ the two-energy density in [5]. Changing variables in (3) gives

$$j_\varepsilon(x,z_1,z_2) = \frac{p_\varepsilon(x+\sqrt{\varepsilon}z_1, z_1)\, p_\varepsilon(x-\sqrt{\varepsilon}z_2, z_2)}{m_\varepsilon(x)} \tag{9}$$

on the angular domain, extended by zero elsewhere. The Jacobians in this formula are one for each pair relative to (x, z_i).

Let φ be the standard Gaussian density and $f(z) = 2z/\sinh(\pi z)$ with its removable value at zero. The exact modular marginal is

$$m_\varepsilon(x) = \varphi(x) e^{\varepsilon/8} \cos(\sqrt{\varepsilon}x/2) F_\varepsilon(\pi + \sqrt{\varepsilon}x) Q_\varepsilon^{\mathrm{im}}(\pi + \sqrt{\varepsilon}x).$$

Here $F_\varepsilon(t) = 1 - e^{-2\pi \min(t, 2\pi - t)/\varepsilon}$ and the image quotient obeys $|Q^{\mathrm{im}} - 1| \le 4e^{-2\pi^2/\varepsilon}$. In particular $m_\varepsilon \le C\varphi$.

Lemma 2. *For every fixed polynomial weight w in $|x|, |z_1|, |z_2|$,*

$$\int w \; |j_\varepsilon - \varphi(x) f(z_1) f(z_2)[1 - \sqrt{\varepsilon}x(z_1 - z_2)]| = O_w(\varepsilon).$$

Proof. Set $L = \log(1/\varepsilon)$ and restrict first to $|x| \le 8\sqrt{L}$, $|z_i| \le 32L/\pi$. This rectangle lies inside the angular domain for small eps. The pair expansion of [5] and the marginal above give a relative $1 + O(\varepsilon P)$ factor, with P a fixed polynomial; the Gaussian ratio is exactly

$$\frac{\varphi(x+\sqrt{\varepsilon}z_1)\varphi(x-\sqrt{\varepsilon}z_2)}{\varphi(x)^2} = e^{-\sqrt{\varepsilon}x(z_1 - z_2) - \varepsilon(z_1^2 + z_2^2)/2}.$$

Its remainder after the displayed linear term is bounded by $C\varepsilon P$ on the rectangle. Integration has no logarithmic loss. Outside, both pair marginals have the global bound $p_\varepsilon \le C\varphi f$ from [5]; each gap marginal thus has exponential tails and every fixed moment uniformly. The middle marginal has Gaussian tails. The union tail probability is $O(\varepsilon^{32})$ times powers of L. Holder gives uniform joint polynomial moments from these marginals; Cauchy–Schwarz then bounds weighted joint tails by $O(\varepsilon^{16})$ times powers of L. The approximating density has the same harmless tails. This proves the lemma without dividing by an uncontrolled small edge marginal. □

5 Time expansion and the surviving neutral memory

For each adjacent pair let $\omega_1 = (e_{\theta_0} - e_{\theta_1})/\sqrt{\varepsilon}$ and $\omega_2 = (e_{\theta_1} - e_{\theta_2})/\sqrt{\varepsilon}$. The exact trigonometric expressions, with $y_1 = x + \sqrt{\varepsilon}z_1$, $y_2 = x - \sqrt{\varepsilon}z_2$, are

$$\omega_i = \frac{-4\cos(\sqrt{\varepsilon}y_i/2)\sin(\varepsilon z_i/2)}{\sqrt{\varepsilon(1-e^{-\varepsilon})}}.$$

On the entire domain, $|\omega_i| \le C|z_i|$ and $|\omega_i + 2z_i| \le C\varepsilon P(|x|, |z_i|)$. These follow from real sine/cosine remainders, allowing polynomial growth. Thus the phase and any fixed number of its time derivatives may be replaced by their $(-2z_1, -2z_2)$ counterparts at $O_{S,k}(\varepsilon)$ cost in Lemma 2. The apparent $\sqrt{\varepsilon}$ term integrates to zero because it is odd in x. Using $\int f(z)e^{-2iuz}dz = g(u)$ gives

$$B_\varepsilon(u, v) = g(u)g(v) + O_{C^k([-S,S]^2)}(\varepsilon). \tag{10}$$

The previous weighted theorem, now used at d=1, gives $C_{1,\varepsilon} = g - \varepsilon R/2 + O_{C^k}(\varepsilon^2)$ and $\widetilde{C}_{1,\varepsilon} = g + \varepsilon R/2 + O_{C^k}(\varepsilon^2)$. Since $s\lambda(1+q) = \varepsilon/2 + O(\varepsilon^2)$, (5) and (10) prove Theorem 1. The new response relative to the trace is

$$\frac{S_\varepsilon(u, v) - C_{1,\varepsilon}(u - v)}{\varepsilon} = R(u - v) + \tfrac{1}{2}g(u)g(v) + O_{C^k}(\varepsilon).$$

The leading neutral addition is a positive rank-one covariance kernel. That describes its time dependence; it is not a new particle or a spatial mode identification.

For the equal mixture $\sigma_{\text{mix}} = (\tau + \sigma)/2$,

$$\sigma_{\text{mix}}(U(u)U(v)) = g(u - v) + \tfrac{\varepsilon}{4}g(u)g(v) + O_{C^k}(\varepsilon^2). \tag{11}$$

The curvature cancellation of [5] therefore does not restore all two-point functions to their leading form. More generally, a constant convex mixture with selected weight a has curvature correction $\varepsilon(a - \frac{1}{2})R_2$, requiring $a = 1/2$ to cancel it for all times; (11) then remains.

Normalizing by the initial selected variance does not remove this memory either:

$$\frac{S_\varepsilon(u, u)}{S_\varepsilon(0, 0)} = 1 + \tfrac{\varepsilon}{2}[\operatorname{sech}^4 u - 1] + O(\varepsilon^2).$$

Thus the effect is not solely a choice of initial amplitude.

6 An observable that sees the odd mixed source

The even two-point function alone does not exhaust the source. Consider the Hermitian energy–slope observable

$$A(u) = \{h, U(u)\}.$$

The microscopic counterpart is the anticommutator of the normalized Hamiltonian and actual exact-pinched slope; the fixed-degree preparation interface of [4] applies. Its trace and energy-only-tilted expectations vanish by U parity, but its selected mean does not:

$$\sigma(A(u)) = -s\sqrt{\lambda}\, H_\varepsilon(u), \tag{12}$$
$$H_\varepsilon(u) = \tau(A(0)A(u)) = \int (a+b)^2 e^{iu(b-a)/\sqrt{\varepsilon}} K_q(a,b)\mu_q(da)\mu_q(db).$$

This follows by keeping $-s\sqrt{\lambda}\{h, U\}$ in (1); the other terms are odd in U after insertion. In particular $H_\varepsilon(0) = 2(1+q)$.

In the center/gap coordinates of [5], the weight is exactly

$$(e_\theta + e_\phi)^2 = \frac{16\sin^2(\sqrt{\varepsilon}y/2)\cos^2(\varepsilon z/2)}{1-e^{-\varepsilon}} = 4y^2 + O(\varepsilon P(|y|,|z|)).$$

It is globally bounded by Cy^2. The weighted density remainder and phase estimate therefore give, with fixed compact-time derivatives,

$$H_\varepsilon(u) = 4g(u) + O_{C^k}(\varepsilon), \qquad \boxed{\sigma(A(u)) = -2\sqrt{\varepsilon}\,\operatorname{sech}^2 u + O_{C^k}(\varepsilon^{3/2})}.$$

This is the odd part of the same preparation memory. It is absent from every energy-only density, not merely from a specific temperature choice. For the equal mixture its coefficient is halved and remains nonzero.

An additional independent selected-transfer check verifies 90 polynomial energy–slope identities; 20 physical time points check the formula, with spectral refinement difference below $4.45\,10^{-15}$. The truncation argument on the following page also applies to H: replace the tilted-pair factor F by $16/(1-q)$ and multiply by $s\sqrt{\lambda}$ for the mean. All saved science is kept separate from the analytic remainder argument.

7 Finite checks and numerical error separation

The new exact checker verifies (4) for every three-block monomial of total degree at most eight, at $q = 1/4, 2/3$ (330 identities). A separate selected open-chord transfer evaluates MU-word $M^\dagger$ for every three-block degree at most six, at two rational (γ, λ) points (168 identities). These are formal algebra checks; the rational parameters need not satisfy $\gamma = e^{-\lambda}$.

For physical q=0.25,0.64, seven time pairs compare a q-Hermite Jacobi spectral implementation with direct angular Pochhammer products (160/224 nodes, 128/192 product terms). The maximum difference is $3.45\,10^{-14}$; quadrature refinement is below $3.05\,10^{-14}$. Four small deformations $\varepsilon = 0.16, 0.08, 0.04, 0.02$ use 28 time pairs and Jacobi dimensions $L = \lceil 12/\varepsilon \rceil + 64$ and $L + 64$. The largest refinement difference is $1.27\,10^{-15}$; the maximum phase-sum condition is 1.118. Exact slices (7)–(8) are also checked.

An independent saved-output bound calculation controls truncation in exact arithmetic, rather than treating refinement as a proof. In the L-node Gauss q-Hermite rule, take

$$m = \lfloor (2L-1)/3 \rfloor, \quad z = \frac{2\max(|u|,|v|,|u-v|)}{\sqrt{\varepsilon(1-q)}}, \quad \eta = \frac{z^{m+1}}{(m+1)!}.$$

Real exponential Taylor polynomials have uniform error at most eta. The q-Mehler map contracts $L^2(\mu_q)$; its finite matrix is orthogonally conjugate to $\mathrm{diag}(q^n)$ and also contracts the Gauss-weighted L^2 norm. For degree at most m the two maps agree at Gauss nodes. The product of three transformed polynomials has degree at most $3m \le 2L-1$, so its quadrature is exact.

Replacing the two exterior exponentials costs at most $2\eta+\eta^2$ in either representation; replacing the middle one costs at most $\eta(1+\eta)^2$. Hence

$$|B - B_L| \le 6\eta + 6\eta^2 + 2\eta^3.$$

For the tilted pair, $m+2 < L$, $2m+2 \le 2L-1$ and $F = 1 - \sqrt{q} + 4\sqrt{q}/(1-q)$ give $|\widetilde{C} - \widetilde{C}_L| \le F(4\eta + 2\eta^2)$. Combining these bounds with (5), the largest refined selected truncation bound is $1.44\,10^{-35}$. Roundoff and angular quadrature discretization are not included in that analytic bound. Product tails are recorded separately.

The largest sampled selected second remainder increases from 0.454 to 0.494 as eps decreases; this finite coefficient is consistent with, but does not prove, the error order. The uniform statement follows from the density proof above.

8 What this adds to the bilocal program

The actual neutral slope is the first coefficient of the neutral memory series entering the reconstructed bilocal. The exact identity (5) supplies its previously missing two-time preparation response in the same microscopic state used for the curvature. It also produces a direct obstruction to selecting a universal leading state by the earlier equal mixture alone.

The next task is to retain the higher neutral residuals N_j and their common-disorder charged cross terms inside the complete bilocal filter [6]. The odd part $-\sqrt{\lambda}\{h, U\}$ is now detected by (12), and must be retained when propagating the full response. No conclusion about all bilocal words, spatial generators, particle masses, finite-G scattering or arbitrary selected interlacings follows from this single neutral coefficient.

All successful previous scientific outputs are reused read-only. The companion evidence file records source and output hashes, dependency versions, both successful LaTeX passes and the eight-page visual check. This stage is local; no new Git or external issue reply is performed.

References

[1] M. Bożejko, B. Kümmerer and R. Speicher, *q-Gaussian processes: non-commutative and classical aspects*, arXiv:funct-an/9604010, Theorems 2.11, 2.13, 4.4, 4.6 and Corollary 4.5: conditional expectations and the classical q-Mehler Markov law.

[2] E. G. Effros and M. Popa, *Feynman diagrams and Wick products associated with q-Fock space*, PNAS 100 (2003), Corollary 2.1 and Theorem 3.2. The q-Wick machinery is prior work.

[3] A. Day and Chengheng, *Single-selected covariant density and curvature dynamics*, 8 November 2026; `manuscript/susskind-slater-selected-state/main.tex`.

[4] A. Day and Chengheng, *Exact energy projection in a single Slater excitation*, 8 November 2026; `manuscript/susskind-slater-orbit-density/main.tex`; with the preceding neutral-slope normalization.

[5] A. Day and Chengheng, *A weighted q-Mehler limit for selected preparations*, 8 November 2026; `manuscript/susskind-weighted-mehler-limit/main.tex`. Its fixed-integer-d theorem is used at d=1 and d=2.

[6] A. Day and Chengheng, *Fixed-charge bilocals and selected preparations*, 8 November 2026; `manuscript/susskind-sector-bilocal/main.tex`.